Sending Cues to TouchDesigner

Gallery Guide Studio: how conversation steps will trigger videos in TouchDesigner

Goal: every message placed in the **Send to TD** column of a playbook is delivered to TouchDesigner the moment the conversation reaches that step, where it triggers a video listed in a TouchDesigner table.

Most of the TouchDesigner side already exists. The earlier **Voiceflow TouchDesigner Installation** project has `trigger-prototype.toe`, which listens for messages on port **9980** and was tested by flashing a rectangle red. This guide reuses that same connection.

1. What kind of connection

An HTTP message from this app's server to TouchDesigner's Web Server DAT. This is the same kind of connection the existing prototype uses: a small web request carrying a cue name, such as `{"cue": "EMERGENCY_VIDEO"}`.

Who sends it matters

The cue should be sent by the app's server on the Mac, not by the phone or the browser:

```
iPhone app --(tunnel)--> App server on your Mac --(localhost:9980)--> TouchDesigner
Computer browser ------->       (same server)
```

- **It works for the phone and the computer alike.** Both already go through the server, so the phone app needs no changes.
- **It avoids the mixed-content problem** flagged in the Voiceflow notes. The phone loads the app over a secure (https) connection, and iPhones block a secure page from contacting a plain, insecure address like TouchDesigner's. Server-to-TouchDesigner traffic isn't subject to that rule.
- **It's fast.** TouchDesigner gets the cue the moment the bot's reply is ready, about half a second before its voice starts.

OSC is the other common way to talk to TouchDesigner and would also work, but HTTP is already built and tested in the existing prototype.

2. What changes in the app

Today, **Send to TD** is only an on/off checkbox in the playbook editor, and nothing reads it. To trigger a specific video, each step's TD cell needs to hold a **cue name**:

- **The TD column becomes text.** Type a cue name such as `EMERGENCY_VIDEO` in FigJam or the playbook editor. A blank cell sends nothing.
- **When the conversation reaches that step** (for example, the bot saying that line), the server sends the cue to TouchDesigner.

- **A TouchDesigner address setting**, defaulting to `http://localhost:9980/trigger`. Change it at school if TouchDesigner runs on a different computer.
- **If TouchDesigner isn't running, the conversation carries on normally.** The cue is simply dropped after a fraction of a second.
- **The computer's chat shows a small marker**, such as `-> TD: EMERGENCY_VIDEO`, so you can see what was sent.

3. Setting up TouchDesigner

Open `trigger-prototype.toe` (or a new project) and build these four pieces.

Step 1: A Table DAT named `video_cues`

Maps cue names to video files. The first row is the header.

cue	file
`EMERGENCY_VIDEO`	`/Users/you/Videos/emergency.mp4`
`WELCOME_VIDEO`	`/Users/you/Videos/welcome.mp4`

Step 2: A Movie File In TOP named `moviefilein1`

This plays the video. Wire it to whatever output you use, such as a Window COMP for the projector. Set its playback so it plays once and holds on the last frame rather than looping, if that's what you want.

Step 3: A Web Server DAT

Use the existing `webserver1`. Set **Port** to `9980` and turn **Active** on.

Step 4: Its callbacks DAT

TouchDesigner creates this automatically next to the Web Server DAT. Replace its `onHTTPRequest` function with:

```
import json

def onHTTPRequest(webServerDAT, request, response):
    body = request.get('data') or b''
    if isinstance(body, bytes):
        body = body.decode('utf-8', 'ignore')
    try:
        cue = str(json.loads(body or '{}').get('cue', '')).strip()
    except Exception:
        cue = ''

    row = op('video_cues').row(cue) if cue else None
    if row:
        movie = op('moviefilein1')
        movie.par.file = row[1].val      # load that cue's video
        movie.par.cuepulse.pulse()       # jump to its start
        movie.par.play = True
        response['statusCode'] = 200
        response['statusReason'] = 'OK'
        response['data'] = 'playing ' + cue
    else:
        response['statusCode'] = 404
        response['statusReason'] = 'Not Found'
        response['data'] = 'unknown cue ' + cue
    return response
```

This script was written from TouchDesigner's documentation and hasn't been run yet. If anything doesn't work, open TouchDesigner's Textport (Alt+T) and send a screenshot of the error.

Step 5: Test it before the app is involved

Run this in Terminal on the Mac. The video for that cue should start:

```
curl -X POST http://localhost:9980/trigger -d '{"cue":"WELCOME_VIDEO"}'
```

4. Two decisions before the app side is built

- **When a cue sits on a BOT step,** should the video start as the bot *begins* that line (suggested), or after it finishes speaking?
- **Should cues on USER steps fire too**, when the caller answers that step, or only on BOT steps?

Once these are settled, the app side gets built: cue names in the TD column, read from FigJam, sent by the server, with the TouchDesigner address setting and the chat marker. Then test end to end with the curl check above.